

Intelligence Graph Ingestion Pipeline

Production Architecture & Technical Design Document | GraphOne / FrontierAtlas

1. Scale Strategy (500,000+ Record Extraction)

Architecting a massive bulk data pipeline capable of acquiring 500k+ entity records without manual intervention or pipeline bottlenecks requires horizontal scalability and decoupled distributed components:

- **Distributed Asynchronous Workers:** Containerized Python asyncio + httpx crawlers deployed on Kubernetes (EKS/GKE) with Horizontal Pod Autoscaling (HPA) triggered by queue depth.
- **Distributed Message Broker:** Apache Kafka or RabbitMQ partitioning target URLs across topic partitions using domain hashing ($\text{hash}(\text{domain}) \% N$), maintaining rate limit isolation per target domain.
- **Proxy Mesh & Anti-Bot Bypass:** Residential proxy pools (BrightData/ScraperAPI) with automatic IP rotation, TLS fingerprint emulation (curl-cffi), and Playwright headless cluster for JS-heavy targets.
- **High-Throughput Storage Ingestion:** Async staging ingestion using PostgreSQL bulk copy (COPY FROM STDIN) achieving > 15,000 inserts/sec.

2. Handling 413 Payload Overflows & 429 Rate Limits

Handling non-deterministic LLM behaviors and gateway throttling across thousands of concurrent extractions:

- **413 Payload Overflows (Intelligent Chunking):** HTML payloads are pre-processed to strip boilerplate markup. Text payloads exceeding context limits undergo sliding-window semantic truncation, preserving title/metadata and dense entity sections while enforcing strict character limits (< 4,000 tokens).
- **429 Rate Limit Resilience (Multi-Tier Fallback Matrix):**
 - Tier 1:** Primary extraction using *Google Gemini 1.5 Flash* (high speed, 1M context).
 - Tier 2:** Fallback to *Groq Llama 3 (Llama-3.1-8b-instant)* on 429 rate limit or HTTP 500 error.
 - Tier 3:** Deterministic rule-based local structural parser (zero-failure fallback guarantee).
- **Exponential Backoff with Full Jitter:** Implements $T_{\text{backoff}} = \min(T_{\text{max}}, T_{\text{base}} \cdot 2^{\text{retry}}) + \text{jitter}$ to prevent thundering herd spikes.

3. Distributed Freshness & Deduplication Architecture

Guaranteeing 24-hour signal freshness for news and job boards while avoiding redundant processing across distributed crawler nodes:

- **Redis Bloom Filters:** In-memory probabilistic data structure ($O(1)$ complexity) for instant check of previously scraped URLs across all crawler pods.
- **SHA-256 Content Fingerprinting:** Hashes normalized body text ($\text{SHA256}(\text{strip_whitespace}(\text{content}))$) to detect syndicated or cross-posted news articles and job duplicate listings.
- **Incremental Timestamp Indexing:** Utilizes HTTP If-Modified-Since and ETag headers along with TimescaleDB hyper-table indexing on `published_date` for rapid 24-hr sliding window queries.

4. Deterministic Entity Resolution Engine

Multi-stage deduplication engine canonicalizing raw organization and product names (e.g., 'OpenAI, Inc.', 'Open AI', 'OpenAI LLC' -> 'OpenAI'):

- **Stage 1: Normalization:** Legal entity suffix stripping (Inc, Corp, LLC, Ltd, Co) and lowercasing.
- **Stage 2: Hash Alias Map:** Instant $O(1)$ lookup against seed dictionary of 50+ known AI canonical entities.
- **Stage 3: RapidFuzz Token Sort Matching:** Token sort ratio comparison with similarity threshold $\geq 82\%$.
- **Stage 4: Entity Mapping Audit Log:** Exports comprehensive log tracking raw name, canonical name, confidence score, and match method.

5. Production Storage Architecture Matrix

Justification and design breakdown for multi-model storage tiering in the GraphOne / FrontierAtlas Intelligence Graph:

Storage Layer	Engine Choice	Architecture Justification & Role
Primary Relational DB	PostgreSQL + TimescaleDB	Enforces ACID compliance, exact schema validation for 5 core entity types, and time-series hyper-tables for 24-hr signals.
Graph Storage	Neo4j / Amazon Neptune	Executes multi-hop graph traversals mapping relationships: (Startup)-[:PRODUCES]->(Product) and (Paper)-[:IMPLEMENTED_BY]->(Repo).
Vector Database	Qdrant / Milvus	Stores high-dimensional dense vector embeddings of research paper abstracts, product features, and job descriptions for semantic RAG search.
Cache & In-Memory State	Redis Enterprise	Provides centralized rate limiting token buckets, distributed locks, Bloom filters, and Celery task state management.

Summary & Compliance

This architecture guarantees high availability, non-hallucinated data lineage, strict 24-hour signal freshness, and seamless theoretical scaling to 500,000+ records across the artificial intelligence and venture ecosystem.